

UniWeb — Approved Feature Set

What to BUILD · How to build · Aadhaar Auto-Mask · Maximum Automation · Explicitly OUT of scope

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Scope of this document

- Only features APPROVED to build for UniWeb P2M platform (technology layer on partner rails).
- No RBI PA license and no PPI license assumed — product stays within that boundary.
- Principle: maximum automation (auto-KYC, auto-mask, auto-agreement, auto-fallback) with safe human fallback when APIs fail.
- Customer full PPI wallet / UPI Scan-and-Pay outward = OUT OF SCOPE (listed at end).

0. Explicitly OUT OF SCOPE (Do not build)

#	Item	Reason
1	Customer PPI wallet (load money, hold balance as PPI)	Requires RBI PPI licence / authorised WaaS
2	Full-KYC customer wallet with ₹2L limit as platform-held funds	Same — regulated PPI product
3	Customer UPI Scan & Pay to any external merchant QR from wallet	Needs sponsor bank UPI + PPI authorisation
4	Storing unmasked full Aadhaar image as the only copy	UIDAI — always store masked; original only if encrypted vault + policy

Aadhaar clarification (IMPORTANT)

- Manual Aadhaar upload IS allowed in fallback flows.
- On every upload (API fetch OR manual file), system MUST auto-mask the first 8 digits in the stored/display image and in any PDF/preview (e.g. XXXXXXXX1234).
- Admin/partner evidence views also show masked Aadhaar by default. Unmasked access only via strict break-glass if ever required by law — not default.
- This is a REQUIRED feature (Section A below), not an out-of-scope item.

A. Aadhaar Auto-Masking (Required on every upload)

Goal: Whether Aadhaar comes from DigiLocker/API or merchant/customer uploads a photo/PDF manually, the platform automatically masks the first 8 digits before persistent storage and before any preview/PDF.

DO ✓ Detect Aadhaar number via OCR on upload; mask positions 1–8 → store masked image + masked number. ✓ Display always XXXXXXXX1234 (last 4 only) in UI, admin, agreement annexure, emails. ✓ Keep audit: original_hash, masked_file_path, ocr_confidence, masked_at. ✓ Apply same rule to e-Aadhaar PDF pages if uploaded as full PDF.	DON'T ✗ Save full visible Aadhaar as the primary file merchants/admins see daily. ✗ Email unmasked Aadhaar to partners by default. ✗ Skip masking when “manual upload” path is used.
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Technical steps

- Upload pipeline: receive file → virus scan → OCR Aadhaar zone → draw mask box / redact pixels on first 8 digits → write masked file to secure storage.
- DB: documents.aadhaar_last4, documents.masked_path, documents.sha256_masked; never put full 12 digits in plain columns.
- Admin Document Manager default thumbnail = masked. Agreement annexure embeds masked version.
- Acceptance: upload sample Aadhaar → UI and stored file show only last 4 readable.

B. Frictionless Auto-Fetch KYC (Maximum automation)

Goal: Merchant onboarding feels automatic — PAN/GST/Aadhaar via API where possible, entity-wise fields, mobile handoff for docs, inline validation, draft save. Admin opens file only when automation fails.

B1 — Entity-wise dynamic form

- First question: Individual / Proprietorship / Partnership / Pvt Ltd / LLP / Trust.
- Show only fields needed for that entity (e.g. CIN+MOA for Pvt Ltd; not for Individual).
- GSTIN present → auto-fetch legal name, address, status from GST API; lock/auto-fill fields.
- PAN → auto-fetch name; compare with GST/MCA where applicable.

B2 — Auto-fetch + Face / Name match

- Providers (config): Karza / Digio / Signzy / similar — swap without rewrite.
- Face + liveness on selfie/video vs ID photo; threshold configurable (e.g. ≥ 88), not fake 100%.
- Name match normalized (Individual vs GST legal name / MCA directors).
- All pass → status=approved by system; trigger agreement eSign (Section D).
- Any fail → action_required with clear reason; merchant retries; after N fails → Manual queue (Section C).

B3 — UX polish (conversion)

- Inline field validation (GSTIN checksum, PAN format, IFSC) before submit.
- Save Draft at every step; resume later.
- Mobile handoff: QR/link “Upload shop photo from phone” while user sits on desktop.
- Progress bar + plain Hindi/English labels (no jargon).
- Capture real client IP + HTML5 geo on each KYC/upload action (Point 1 legacy).

DO ✓ Auto-approve only when API+face+name all pass. ✓ Block live collect until approved OR admin manual approve after fallback. ✓ Store every provider request id for evidence pack.	DON'T ✗ Let merchant take live payments while KYC = pending with no limit. ✗ Require admin to open every file when APIs already returned Verified.
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C. Auto-KYC Fail → Manual Fallback (Admin Document Manager)

Goal: When API/OCR/face fails, merchant is not stuck forever. Admin gets a clean queue, can view masked docs, approve/reject with reason, and optionally forward to partner.

- Queue: merchants with status=action_required or manual_review, sorted by waiting time.
- Document Manager: list all files (masked Aadhaar), zoom, verify checklist, Approve / Reject + note.
- On Approve: same as system approve → agreement flow if not signed → ready for partner forward.
- On Reject: merchant notification with what to fix; resubmit allowed.
- Never “connect live keys” without Approved status (system or admin).

D. Merchant Agreement PDF + Aadhaar eSign + Distribution

Goal: After KYC approved, generate legal agreement PDF (T&C, Privacy, Refund, Compliance + partner names for this merchant + KYC annexures masked) → merchant scrolls → Accept → Aadhaar eSign (Digio/Leegality) → embed signature → email merchant + notify partner.

DO ✓ Dynamic PDF: company clauses + injected gateway names (Cashfree/Razorpay/...) for this merchant. ✓ Annexures: masked Aadhaar, PAN, GST, photo, shop, etc. at end of PDF. ✓ Stamp on acceptance pages: IP, lat/lng, accuracy, timestamp, device. ✓ Sequential: scroll policies → I Accept → trigger eSign API → Complete only after eSign success. ✓ Auto-email signed PDF to merchant; push copy/metadata to partner API or compliance email.	DON'T ✗ Mark agreement complete without eSign when eSign is enabled for that flow. ✗ Attach unmasked Aadhaar in annexure. ✗ Skip IP/geo stamp on signed copy.
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Technical steps

- Template engine (HTML→PDF or DOCX→PDF) with variables: merchant legal name, PAN, GSTIN, partner_list[], annexure_files[].
- eSign adapter: create document → redirect/OTP Aadhaar → webhook signed → download signed PDF → store.
- agreement_records: status draft|accepted|signed|distributed, ip, geo, signed_at, file_path.
- Acceptance: test merchant completes eSign; PDF shows mask + IP/geo; email received.

E. Transparent Checkout + Merchant White-Label

Goal: Customer checkout is clean, trustworthy, white-labeled to merchant brand; no hidden charges; T&C; links; trust badges; smooth UPI/card/netbanking without full page reload where possible.

E1 — Transparency

- Show amount + fee + tax breakdown before Pay.
- Checkbox + links: Terms, Privacy, Refund above Pay Now.
- Badges: Secure / UPI Safe (accurate claims only — do not claim PCI if not in scope).
- Methods with icons; UPI → QR/intent inline.

E2 — White-label builder (merchant)

- Merchant settings: logo, primary/secondary color, brand name → applied on hosted checkout.
- Optional premium: custom subdomain mapping (pay.merchant.com) — DNS + TLS provision.
- Success/Failure page: custom message, support contact, redirect URL.
- Toggle/order payment methods per merchant (within what platform enables).

Global UX rule: All portals (Admin/Staff/Merchant/Customer) — simple words, wizard for long forms, large touch targets, mobile-first. Prefer “Send money” over “Initiate disbursement”.

F. Multiple P2M QR (Fixed + Open Amount + Labels)

- Merchant creates many QRs: label (Counter 1, Delivery, Campaign), mode fixed|open, amount if fixed.
- Fixed = amount locked in UPI intent; Open = customer types amount.
- Each payment tagged with qr_id; reports split by label.
- Disable QR without deleting history.

G. PAN vs Multiple GSTIN (Multi-business, one login)

Scenario	Action
Same PAN, no GST, signup again	BLOCK — account exists, please login
Same PAN + same GSTIN	BLOCK — business already registered
Same PAN + different valid GSTIN	ALLOW — new merchant under same user; business switcher
GSTIN format	Positions 3–12 must equal PAN; checksum validate

Model: users (login) → many merchants (pan+gstin) via user_merchant_roles; session active_merchant_id.

H. Developer Portal + Sandbox API Docs

- Clean docs site: Collections, VA, Payouts, KYC helpers, Webhooks, Errors.
- Each endpoint: plain English purpose + request + response + PHP/Node/Python/Java/cURL snippets.
- Merchant dashboard: generate Sandbox keys instantly; test pay + test webhook.
- Webhook signature verification guide; error code table with simple meanings.

I. Money Safety (Mandatory with P2M live)

Not optional if real money flows. Build with mocks first; plug partner keys later.

Item	Rule
Webhook idempotency	UNIQUE event_id; claim-first; one credit only
Ledger states	available → in_transit → settled reversed
Beneficiary	Penny-drop/verify before auto payout; else Manual UTR only
Auto payout path	Adapter ready; Manual UTR emergency fallback
Fail payout	Reverse to available + clear reason to merchant

J. Approved for LATER (not blocking P2M launch)

Item	When
Full e-Mandate (UPI Autopay + e-NACH + debit card + Aadhaar eSign NACH)	After collect+settle stable
Dynamic MDR engine + public pricing from admin	When pricing changes often / many merchants
Multi-gateway KYC forward matrix	When 2+ live partners
Refund: 24h choice Wallet credit (closed refund-only) vs bank	Only closed-loop refund credit — NOT PPI load/UPI-out

Refund wallet note

- If you ever offer “keep refund in wallet”, it must be closed-loop credit for future pay-ins on UniWeb merchants only — no customer top-up, no Scan & Pay to external UPI.
- Full PPI / outward UPI remains OUT OF SCOPE until licence + bank WaaS.

K. Recommended build order

Order	Block	Outcome
1	A Aadhaar mask + B Frictionless KYC + C Manual fallback	Onboard without compliance holes
2	D Agreement + eSign	Legally signed merchants
3	F Multi QR + G Multi-GST + I Money safety	Real P2M collect + safe ledger
4	E Checkout white-label + transparency	Higher conversion / trust
5	H API docs + sandbox	Developers integrate
6	J Later items	Growth features

L. Acceptance checklist (QA)

- Manual Aadhaar upload → stored + UI show only last 4 digits; first 8 masked in image.
- Proprietorship vs Pvt Ltd forms show different required fields; GST auto-fill works on sandbox provider.
- API fail path lands in Admin Document Manager; approve moves merchant to Approved.
- eSign completes → PDF has partner names + masked annexures + IP/geo stamp; email received.
- Two GSTINs same PAN → one login, two businesses, switcher; duplicate PAN+GST blocked.
- Three QRs (2 fixed 1 open) → payments report by label.
- Webhook replay twice → single wallet credit.
- Checkout shows fee breakdown + T&C; checkbox; merchant logo/colors apply.

Final principle

- Automate everything that APIs allow (KYC fetch, mask, eSign, credit, payout path).
- When automation fails → clean manual fallback — never silent stuck, never unapproved live.
- Never build customer PPI / external UPI Scan-and-Pay until PPI + bank authorisation exists.
- Masked Aadhaar is default everywhere; that is a feature you asked for and it is in scope.